4

Ventilation

1.0 BACKGROUND

Semiconductor manufacturing facilities intensively use the movement of ultraclean air to control the levels of particulates in their fabrication areas. Ventilation is one of the more important contamination control technologies used in fab processing, but only one of the twelve major components identified in Fig. 4.1.[1]

The movement of large volumes of air throughout the fabrication area creates both beneficial and potentially negative impact on the exposure potential of employees working in the fab. Specifically, the large volumes of high velocity air moving within the fab environment can create in excess of sixty air changes per hour, and act as a major source of dilution ventilation for any hazardous emissions that may get released into the fab environment.[2] Figure 4.2 provides a graphical categorization of the number of fresh air changes per hour of the general ventilation systems in fabs that were studied by University of California at Davis.[3]

It is very common for minor releases of compounds with low odor thresholds to be quickly spread within the air handler zone servicing that area of the fab. This generalized odor results in the potential for large area evacuations for an "unknown odor." In fact, odor identification is one of the roles the industrial hygienist performs in responding to emergencies, and evacuations. For this reason, some companies have opted for the installation and use of continuous gas monitoring systems to aid in the definition of unknown odors. See the specific subsection in Ch.3: Chemical Agents—

Continuous Monitoring for additional information. These monitoring systems are useful for determining the airborne concentrations of toxic, corrosive, and organic materials from leaks, spills, effluents from equipment, and releases during equipment upsets or maintenance. Odor identification is of such importance to the industrial hygienist that a subsection is included in Ch. 6, Indoor Air Quality, subsection 6.2, Odor Identification.

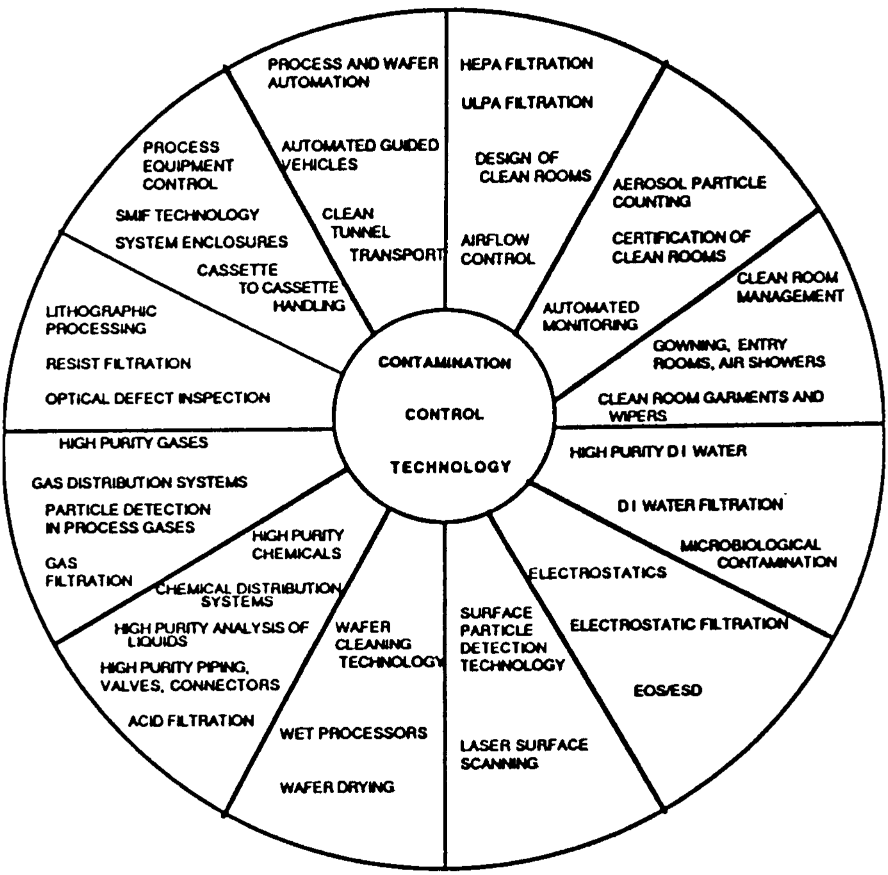

Figure 4.1. Contamination control technologies used in semiconductor processing.[1] *(© 1988, Noyes Publications, reprinted with permission.)*

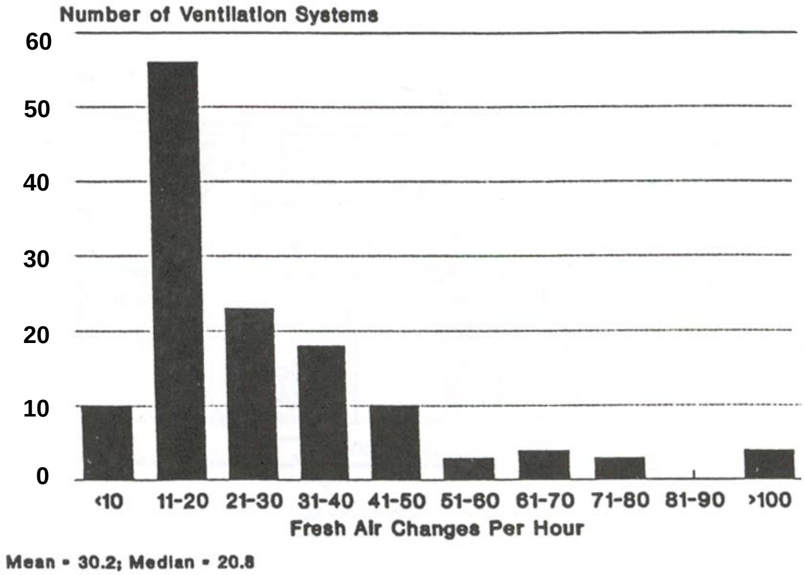

Figure 4.2. Graphical categorization of the number of fresh air changes per hour of the general ventilation systems in fabs studied by Univ. of Cal. at Davis.[3]

The use of direct reading instruments for evaluation of transient odors is also another area of industrial hygiene support. Portable instruments that detect the compounds typically encountered in the fab are used very frequently, as well as ventilation monitoring equipment. See the odor identification subsection in Ch. 6 for additional information on monitoring techniques and equipment.

Within the fab environment, a competition for air is going on due to the enormous volumes of air being moved in, out, and within that space. Part of the air is being supplied to create a positive static pressure within the fab area to prevent the infiltration of dirtier air from the exterior. There are also large volumes of air being recirculated within the various zones of the fab, and then very large quantities of air being exhausted from process equipment or tools that create heat, off-gas materials or decompose chemical species that are part of the process technologies being used. Figure 4.3 provides a schematic representation of the airflow characteristics in a fab area with a laminar flow ceiling, and return air through the raised floor and service chase.[4] Figure 4.4 identifies the relative proximity of the fab area proper to other ancillary areas such as gowning areas and service aisles used for equipment maintenance or chemical pass-through.[5]

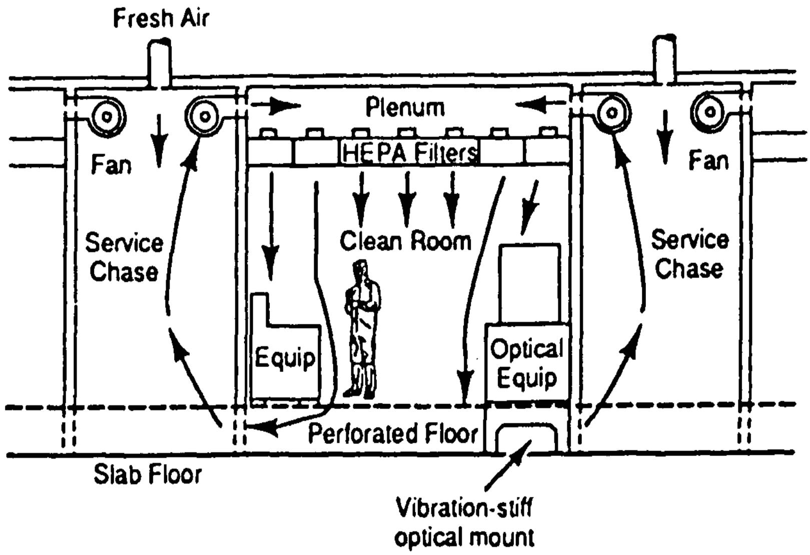

Figure 4.3. Schematic representation of airflow characteristics in typical fab area.[4] *(Van Zant, P.,* Microchip Fabrication, *© 1990, McGraw-Hill, reprinted with permission.)*

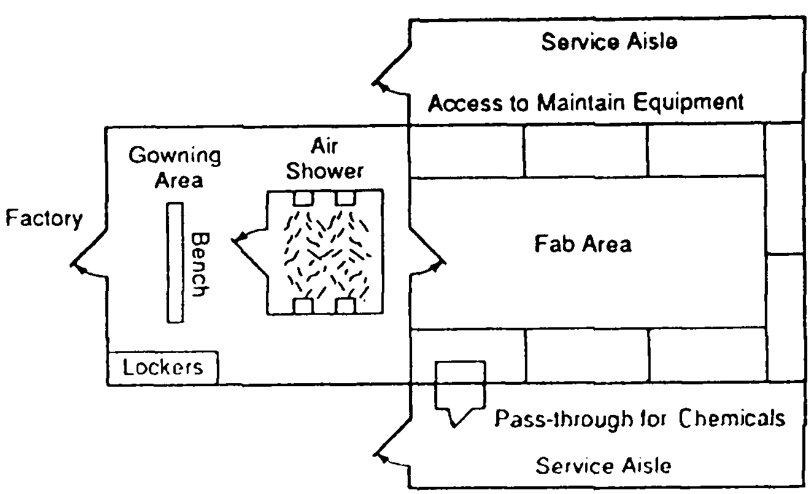

Figure 4.4. Schematic of the relative proximity of the fab area proper to the other ancillary areas.[5] *(Van Zant, P.,* Microchip Fabrication, *© 1990, McGraw-Hill, reprinted with permission.)*

The facilities engineering groups at the site are constantly forced to recalculate the quantities of air being supplied, recirculated, and exhausted to ensure that a positive pressure is maintained within the fabrication process bays. In fact, the areas that require the highest levels of particulate control, such as the photolithography, deposition, and implant areas, are kept at a positive pressure in relation to the adjacent process bays that are not as particle sensitive.[6][7] These tight control requirements over a typical fab area, process chemicals, and deionized water are characterized in Fig. 4.5.[8]

Nose bleeds in the occupants of older fabrication areas have been associated with low humidity and high air flow. However, this is not an issue in newer fab areas where humidity is added for process reasons (e.g., to maintain resist sensitivity in a narrow range and control electrostatic discharge potentials).

SOURCE	SPECIFICATION
Air	
Particulates LSI	Class 100 @ 0.5 micron
VLSI	Class 10 @ 0.3 micron
Humidity	15-50%
Temperature	68-74°F
Photochemical Smog	2 pphm (parts per hundred million)
Deionized Water	
Resistivity	15-18 meg Ω cm
Particulate	Less than 100 particulates per centiliter after 0.5 micron filtration
Bacteria	Less than 100 colonies per ml sample after 0.5 micron filtration
Chemicals	
Metallic Impurities	Less than 1 ppm
Filtration	To 0.2 micron
Gases	
Purity	Greater than 99.9%
Water Vapor	Less than 5 ppm
Filtration	To 0.3 micron
Static Charge	Less than 50 volts

Figure 4.5. Summary of typical clean-room requirements for air, water, chemicals, gases, and static charge.[8] *(Van Zant, P.,* Microchip Fabrication, *© 1990, McGraw-Hill, reprinted with permission.)*

2.0 TYPES

There are typically three distinct types of air movement systems in use in most semiconductor fabrication areas: *general HVAC, local exhaust ventilation* (LEV) system, and then dedicated *particulate control systems* (PCS) that are located at the individual piece of equipment or workstation(s). Figure 4.6 provides a visual representation of a typical "generic" ventilation system that identifies both the general HVAC component, as well as the local exhaust ventilation system.[9] Figure 4.7 provides a more detailed view of the third component of this air movement system, namely the particulate control system which may encompass an enclosure around an individual piece of process equipment, an area of dedicated process equipment (typically called a bay), or combination of different types of equipment in a zone of the fab, such as the photolithography area.[10]

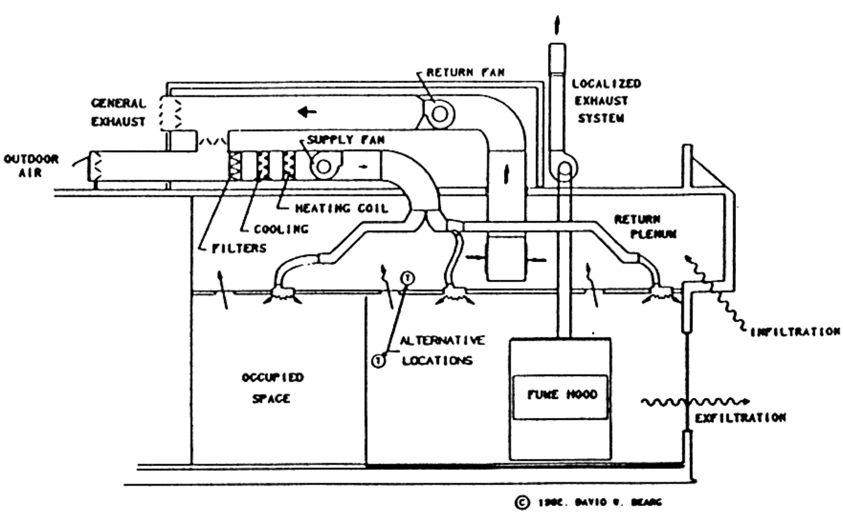

Figure 4.6. A typical "generic" ventilation system that identifies both the general HVAC component and local exhaust ventilation system.[9] *(Reprinted with permission from* Indoor Air Quality and HVAC Systems. © *Lewis Publishers, a subsidiary of CRC Press, Boca Raton, Florida)*

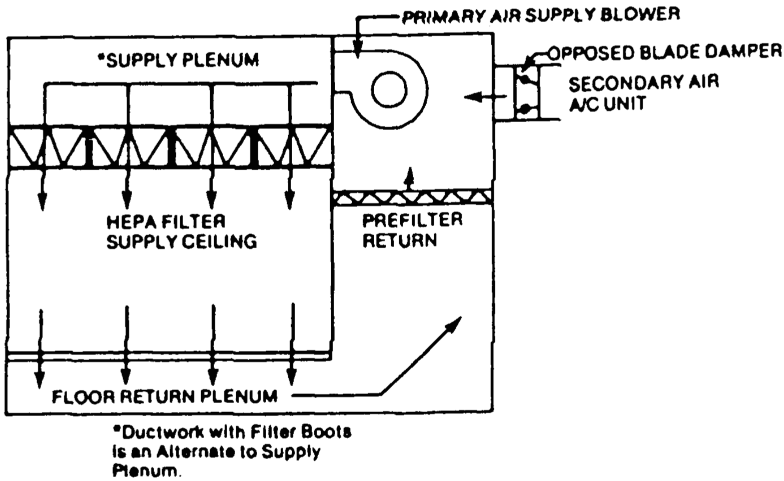

Figure 4.7. A diagram of a particulate control system (raised floor cleanroom).[10] *(© 1988, Noyes Publications)*

A fourth level of particulate control involves actually putting the fabrication worker in totally enclosed "bunny suits" that have air supply and filtration systems to control the levels of particulates generated by the workers in the fabrication area. The human occupants of the fabrication areas are considered very potent generators of fine particulates from their exhaled air, shedding of skin/hair, and from their clothing/shoes. Figure 4.8 identifies the relative contribution of various wafer fab particulate sources. As can be seen, the equipment and people within a fab area contribute in excess of 50% of the total particulate contamination in a fab area.[11] This realization is the driving force behind the isolation of the worker from the work environment using *wafer isolation technology* (WIT) and the pervasive use of automated equipment loading and wafer transport systems in most of newest generation fabrication areas.[12][13] Table 4.1 provides an overview of the terminology and environment size in wafer isolation technology (WIT).[12]

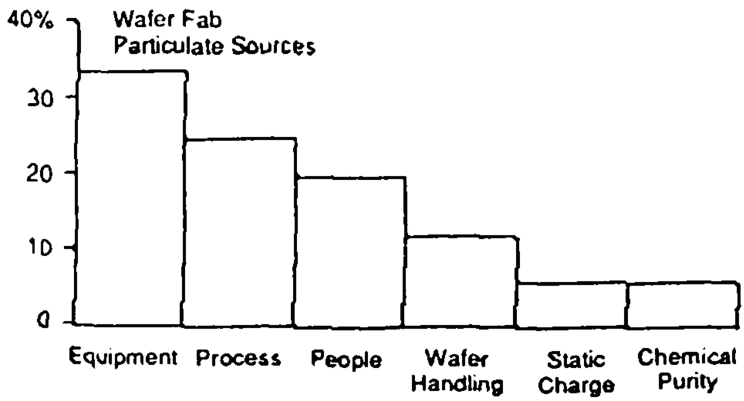

Figure 4.8. The relative contribution of various wafer fab particulate sources.[11] *(Van Zant, P.,* Microchip Fabrication, *© 1990, McGraw-Hill, reprinted with permission.)*

Table 4.1. Overview of the terminology in Wafer Isolation Technology (WIT).[12] *(© 1993,* Semiconductor International, *used with permission.)*

Wafer environment size	
Maxi-	**Cleanroom:** a facility in which the air, construction materials and operating procedures are regulated to control airborne particles.
	Minienvironment system: a tool enclosure, In/Out device, wafer carrier enclosure. **Minienvironment:** an enclosure around a process/support tool that isolates it from room contamination and usually provides the tool with its own supply of conditioned air. **MAP minienvironment:** minienvironment with a manual access port fitted with arm length stretch gloves. **Microenvironment** describes the following: **Cluster tool:** two or more independent process modules integrated by a central robotic wafer handler in a vacuum isolated environment. **Tunnel:** an enclosed track for the transport of single wafers or cassettes between processes. **Run Box:** an enclosed carrier for wafers and cassettes. **SMIF-Pod:** *an environmentally secure wafer container used for storage and transfer of wafers.*
Micro-	

2.1 General HVAC

The general HVAC is responsible for the essential ventilation supplied to each of the occupied areas within all office and production buildings. This is generally described as the HVAC (heating, ventilating and air conditioning) system. Its primary function is, as the abbreviation HVAC would connote, providing ventilation to occupied spaces that are necessary to control the temperature, humidity, particulate levels, and general levels of contaminants being liberated from the occupants, the furnishings, and to provide comfort ventilation in the form of air movement.

Many typical *indoor air quality* (IAQ) or *sick building syndrome* concerns have become more common because of energy efficiency requirements resulting in lower levels of fresh air or make-up air being provided to the general HVAC system. Numerous code bodies specify minimum rates of air movement per occupant (cubic feet/minute per occupant—CFM/occupant) to counter the adverse effect of energy efficiency efforts. Typically for interior spaces, ASHRAE 62-1989 recommends a minimum of 20 CFM of outside air per occupant in office areas.[14] Fabrication area HVAC systems are unusual for a variety of reasons: the acceptable levels of temperature, humidity, static pressure, particulates, and oxidants are much more tightly controlled than anything needed from a comfort or code compliance perspective (see Fig. 4.5 for specific levels in a typical fab).

Cleanroom/Fab Area Classifications. U.S. Federal Standard 209E, titled the Federal Standard, Clean Room and Work Station Requirements, Controlled Environments,[15] establishes standard classes of air cleanliness for airborne particulate levels in cleanrooms and fabrication areas. This standard is issued by the General Services Administration of the United States government. The standard was originally designed for use by federal agencies, but it has been adopted by U.S. industry, and is used world-wide. It prescribes methods for class verification and monitoring of air cleanliness. It also addresses certain other factors that affect control of airborne contaminants. The federal standard however, does not address the occupational health concerns of employees working in cleanroom or fab environments. This system categorizes cleanrooms or areas into classes based on the statistically allowable number of particles, greater than or equal to 0.5 micrometers in size, per cubic foot of air. The cleanroom or fab area classes are shown in Table 4.2.[16]

Table 4.2. Class Limits in Particles per Cubic Foot of Size Equal to or Greater than Particle Sizes Shown.[16] *(© 1992, ACGIH, used with permission.)*

	Measured Particle Size (micrometers)*				
Room Class	**0.1**	**0.2**	**0.3**	**0.5**	**5**
0.01	0.35	0.075	0.03	0.01	NA
0.1	3.5	0.75	0.3	0.1	NA
1	35	7.5	3	1	NA
10	350	75	30	10	NA
100	NA**	750	300	100	NA
1000	NA	NA	NA	1000	7
10000	NA	NA	NA	10000	70
100000	NA	NA	NA	100000	700

* The class limit particle concentrations are defined for class purposes only and do not necessarily represent the size distribution to be found in any particular situation.

** Not Applicable.

As a method of comparison, an outdoor environment would be greater than Class 500,000; a typical room in a house approximately Class 100,000; and a semiconductor assembly area approximately Class 10,000.[17] Most of the older existing fab areas are Class 1000 to Class 100, with the newer fabrication areas having much tighter levels of control for temperature, humidity, air velocity, and airborne particulates. These new fab areas have airborne particulate levels that are Class 10, Class 1, or even Class 0.01 (typically in minienvironments). These requirements for tighter contamination control in the fab have resulted in lowered acceptable particle counts, and secondarily, a focus on specifications for conditioned air, e.g., temperature control (± 0.5°C), humidity (± 1% or better) and velocity (variation ± 10%). The size and number of particles that are acceptable for fabs producing 0.4 to 0.8 µm devices today will become totally unacceptable as finer line widths are needed for the next generation of semiconductor devices. Table 4.3 provides a fab environment comparison of a Class 1 cleanroom, minienvironment, and requirements for 64 and 256 Mbit devices.[18]

By applying somewhat arbitrary assumptions for dust density (2.5 g/cm^3) and mass median diameter (1.5 µm), a particulate count can be converted to an approximate mass concentration. Using this approach suggests a ratio of 6000 particles per cubic foot to 1 µg/m^3 of respirable dust.[19] When compared to the cleanroom classifications typical for semiconductor manufacturing (i.e., Class 100 and less), this indicates particulate exposure in the fab areas is not an issue even for particulates with low TLV®/PELs such as arsenic, chromium, and lead. However, this is not always the case for certain maintenance operations where localized particulate exposures can occur.

Table 4.3. Fab environment comparison of a Class 1 cleanroom, minienvironment, and requirements for 64 and 256 Mbit devices.[18] *(Reprinted with permission.)*

Fab environment comparison

Environmental consideration	Class 1: Based on audits of operational fabs	Minienvironments with mechanical I/O	64 Mbit (proposed specs) per SEMATECH/JESSI	256 Mbit (proposed specs) per SEMATECH/JESSI
Airborne wafer environment (per Fed. Std. 209E)	Class 1-10 (particle bursts)	Class 0.01-0.1 (no particle bursts)	Class 0.1	Class 0.01
Atmosphere type	Air	Air/N_2	Air/N_2	Air/N_2
PWP (at ≥ 0.2 μm, 200-mm wafers)*	0.5-1.0	0.005-0.04	0.3	0.06
Storage/transport environment	Air	Air/N_2	Air/N_2	Air/N_2

*Particles added per wafer pass, which includes opening a cassette container, transferring the cassette to a process tool, and reloading the cassette in the container.

High Efficiency Filters. In order to meet the class limits, a *high-efficiency particulate air* or *attenuator* (HEPA) filter or an *ultra-low penetration air* or *attenuator* (ULPA) filter is required. A HEPA filter is a disposable, extended-media, dry-type filter in a rigid frame with a minimum particle collecting efficiency of 99.97% for 0.3 micrometer, thermally generated dioctylphthlate (DOP), or specified alternate, aerosol particles at a maximum clean resistance of 1.0" water guage when tested at rated air flow capacity. An ULPA filter is a disposable, extended-media, dry-type filter in a rigid frame with a minimum particle collecting efficiency of 99.999% for particulate diameters greater than or equal to 0.3 micrometer in size. ULPA filters exhibit 99.9995% efficiency on particles in the 0.10 to 0.20 micrometer range.[16] These filters are used extensively in semiconductor processing for cleaning the air being supplied to or recirculated to the fabrication area. The ability to control airborne particulates to these amazingly low levels is integral to semiconductor fab processing.

Manufacturers rate HEPA and ULPA filters according to standards adopted by the Institute for Environmental Standards (IES) and Military Standards. IES RP-CC-001-88-T specifies the procedures and testing equipment required by Military Standard 282 DOP, a particle size efficiency test that uses a cloud of uniformly sized 0.3 micrometer dioctylphthalate (DOP) droplets. Efficiency of the HEPA filter is determined by comparing the downstream and upstream particle counts using % DOP efficiency. To be designated as a HEPA filter, the filter must be at least 99.97% efficient, i.e., only three particles of 0.3 micron size can pass for every ten thousand particles fed to the filter. The testing and certification protocols are specified in ASHRAE Standard 52-76 Test Method. Efficiencies are categorized into four separate groups, with group IV filters defined as extended-area pleated HEPA-type filters of wet-laid ultra-fine glass fiber. Biological grade air filters are generally 95% DOP efficient; HEPA filters are 99.7 and 99.99%; and ULPA filters are 99.999%. HEPA filters are used in a variety of configurations depending on their placement in the system, e.g., in-line in the main air handling units, as "laminar flow ceilings or floors," or in combination with the Particle Control System (PCS) at the process equipment/tool or workstation. HEPA filter technology is at the heart of all HVAC systems or subsystems within the fabrication area environment. It should be noted that newly installed HEPA filters themselves, used in PCS systems, are known to be potential sources of odors in the fab area due to offgasing of the liquid gel sealants used within the HEPA filters.[20]-[23]

The overall performance of the cleanroom or fab area is characterized in testing methods specified in IES-RP-CC-006-84. This standard defines terms having special meaning and describes test procedures to assure proper cleanroom operation. Uniform air flow is defined as unidirectional with all velocity readings within 20% of the average velocity of the work area. The air velocity at the work area is generally about 100 fpm; however, design conditions may require velocities of 10 fpm or lower.[24]

IH Considerations. The industrial hygienist must be constantly aware of the important pressure differentials in the numerous zones or areas within the fab. These differentials are controlled by the relationship of the general HVAC system, the local exhaust ventilation (LEV) systems, and sub-level particle control systems (PCS) at the process tool or bay within the fab. It is imperative that the industrial hygienist understand the specifics of each of the systems prior to performing LEV system measurements, laying out gas detection system monitoring line placement, investigating odors or transient “smells”, or specifying minimum exhaust velocities or volumes on potentially hazardous processes or procedures. It’s beyond the scope of this chapter to provide a definitive description of the general HVAC system. Instead, the focus will be on the LEV and PCS systems that act as the primary engineering controls that provide protection for airborne hazards to the occupants of the fab areas.

Special mention is needed regarding the extensive use of vacuum processing in the fab. These systems and their ancillary piping/plumbing systems are notorious sources of transient odors, and brief discharges. Maintenance activities and pump/exhaust system blockages are fairly common occurrences that dictate strict precautions in the timing of routine maintenance, use of proper protective equipment and potential localized control of occupied areas adjacent to the pumps, cold traps, etc. Another alternative is to remove the pump system and do the preventive maintenance (PM) in a ventilated hood.

Emissions from sources external to the fab are a fairly common occurrence, such as diesel exhaust from a delivery truck idling at a delivery dock or pad; cafeteria grill exhaust odors; or maintenance activities taking place (e.g., welding, spray painting, installation of carpeting or floor tiles, or emissions from adjacent cooling towers that are being dosed with biocides or ozone), and then being picked up by the intake of the general HVAC system. This category of unknown odors in fab areas or other occupied areas of the buildings require the devoting of industrial hygiene resources to determine the “cause” and “clear” the affected area for reoccupancy. These issues should also be considered in the design, layout, and construction of new facilities.

Because of the awareness of fab employees to the wide variety of chemicals used in the semiconductor processes, odors from external sources can to trigger precautionary evacuations of the fab or adjacent areas. These evacuations are expensive in terms of downtime of the fab and the uncertainty they create in the fab employees regarding the safety of the work area (which directly affects employee morale). Therefore, a good deal of time and effort is spent on preventing these odors, and when they occur, identifying and eliminating their root causes.

The location of the intakes for these high volume, high velocity general HVAC systems is critical, in relationship to the LEV system exhausts, pad areas, roadways, adjacent businesses or operations, and facility support systems. Due to the frequency of these odor occurrences, or re-entrainment of odors or hazardous emissions from adjacent LEV systems, there is a growing trend toward the modeling of airflow patterns for general HVAC system intakes and LEV exhaust systems (such as wet scrubber exhaust or exhaust conditioning stacks). Figures 4.9 and 4.10 provide a two and three dimensional representation of the recirculation patterns around buildings. These representations are not intended to provide quantitative data on LEV stack placement, but give a general idea of the effects that can be anticipated.

As a secondary line of protection for the removal of odors that may be drawn into the air intakes, the installation of carbon adsorption filters on the intakes to the general HVAC systems may help to alleviate these transient odors, and to control the levels of *volatile organic compounds* (VOCs) in the fabrication areas. Carbon adsorption is not a panacea for the removal of all unwanted odors, and it relies on physical adsorption of the chemical species on a carbon bed or cannister. Many compounds will pass through carbon unless it has been pretreated or specifically chosen to adsorb the target chemical species and has sufficient residence time as it travels through the carbon bed. Even with the best match of carbon to the potential gas or vapor that may be drawn into the general HVAC system, carbon has a fixed capacity and is greatly affected by relative humidity and flow rate. It also requires periodic maintenance and change-out of the carbon which is expensive and time consuming.

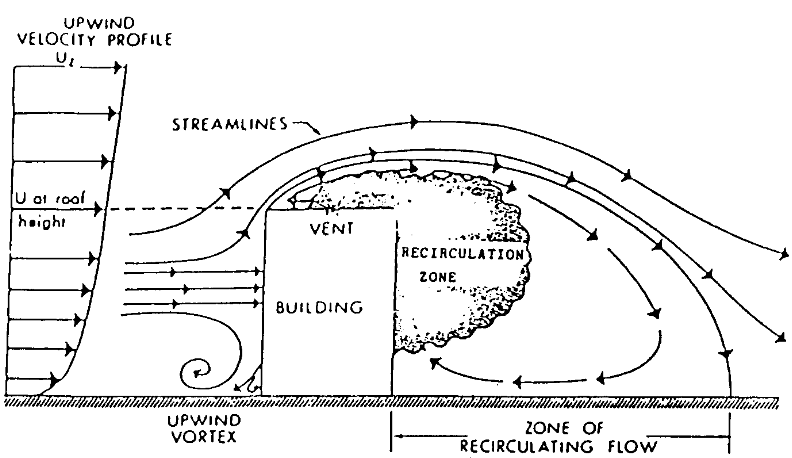

Figure 4.9. Two-dimensional recirculation pattern around a building.[25] *(Reprinted with permission from* Indoor Air Quality and HVAC Systems. *© 1993, Lewis Publishers, a subsidiary of CRC Press, Boca Raton, Florida.)*

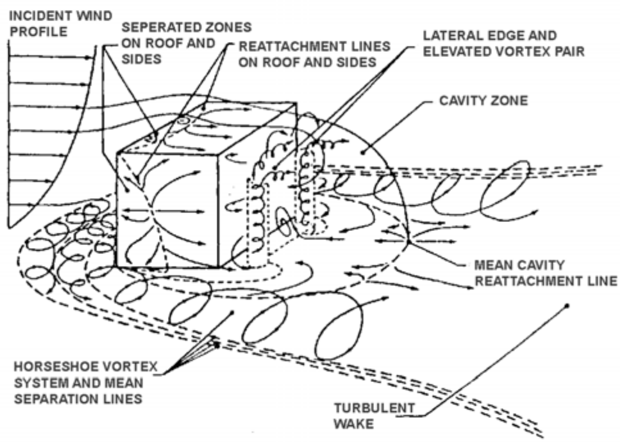

Figure 4.10. Three-dimensional recirculation pattern around a building.[26] *(Reprinted with permission from* Indoor Air Quality and HVAC Systems. *© 1993, Lewis Publishers, a subsidiary of CRC Press, Boca Raton, Florida.)*

2.2 LEV Systems

The second category of ventilation systems within the semiconductor fabrication area is one familiar to all practicing industrial hygienists and facilities department HVAC engineers—the LEV systems. The most unusual aspects of the LEV systems in use in this industry are the numbers and diversity of configurations, properties of the materials that are exhausted, and the number of chemical species that may be exhausted within the system. The very real challenge to the industrial hygienist is understanding the dynamics of the process equipment or tool, the initial chemicals running in the process, the decomposition products that are formed, their interaction in the LEV system, and ultimately the conversion of these species into environmentally-acceptable residues or discharges.

The complexity of this topic necessitates a high level approach. An overview of the various types of LEV systems commonly in use in the industry is provided, with unusual hazards or experiences noted for each type. LEV systems are designed to create a zone of negative pressure around the intended effluent source, capture the intended material, and safely transport the hazardous, toxic, or particle-generating source to an appropriate treatment system, or through dilution to the ambient outdoor environment. Finally, a more extensive listing of processing technology steps, chemicals in use, and potential hazards is provided.

Types of LEV Systems. Table 4.4 illustrates the various types of LEV systems commonly found in the semiconductor industry for the exhausting of hazardous materials. It is not intended to be a totally comprehensive list, but rather a general indicator of the many categories of LEV systems and unique hazards of the semiconductor industry. It must be noted that the majority of these systems can easily be impacted by the local particle control systems (PCS) that may be provided above (*vertical laminar flow*—VLF hoods), in front of the workstation (*horizontal laminar flow*—HLF hoods), or as an add-on physically to the piece of equipment *(minienvironment).* Air movement from the laminar flow hood (VLF/HLF) disrupts the zone of capture for the LEV system. Figure 4.11 provides a cross section of a VLF fume exhaust hood illustrating the effect of the VLF on the hot etch pot LEV exhaust flow in a standard wet sink design. Alternatively, if a positive pressure or push is added adjacent to the LEV intake point (through overhead VLF hoods or positive static pressure in the fab zone relative to the LEV), a form of push-pull system may be created. This potential for affecting the LEV necessitates careful consideration of the VLF or HLF hood system, laminar flow ceiling, or minienvironment on the LEV system. Further mention is made of this phenomenon in the ventilation monitoring section of this chapter.

Table 4.4. Types of LEV Systems Commonly Used in the Semiconductor Industry.[1]-[4][9]

Laboratory-Type Hoods. Used in many locations for storage/use of chemicals, parts storage, or in QA/QC/Reliability labs for analytical procedures; ensure materials of construction are compatible with starter and intermediate chemicals, and sash adjustments are marked for minimum face velocities of 80–120 lf/min as per ASHRAE/ANSI Z9.5-1992, Lab Ventilation; crossdrafts from general HVAC and especially PCS/VLF can disperse air contaminants, as well as foot traffic in front of hoods; pay attention to compatibility of materials stored in the hood.

Table 4.4. (Cont'd)

Wet Sinks (primarily for etching or cleaning of wafers/boats). Plenum exhaust with shroud that fits over the deck surface to allow better capture at the etch/clean baths and to direct overhead vertical flow from laminar flow hood (making a push-pull system), with face velocities at shroud of 100–125 lf/min, in addition to compatibility of materials in the sink; pay attention to their compatibility with drains (e.g., cyanide solutions should not be kept in sinks with acid aspirators). Another approach used within the industry is to provide open wet sinks without shrouds with an exhaust volume of 125–150% of the laminar flow supply.

Gas Cabinets. Use 150 CFM per bottle in cabinet design; UFC Article 51.107(c)3 requires gas cabinet be ventilated with 200 FPM average face velocity at the access port, with minimum velocity of 150 FPM at any point; ensure vent lines are routed to gas conditioning system or silane burn-off system; make sure exhaust from one cabinet is not intermingled with other cabinet exhausts unless chemically compatible.

Gas Jungles or Gas Control Valves. Must be exhausted; some companies have set internal standards, for example, a minimum of 5 air changes per minute (ACM) and 125 feet per minute of face velocity if accessible, or 5 ACM if not accessible.

Open Surface Tanks (plating or degreasing). Commonly used in "back-end" processing for adding precious metal layers; depending on the plating tank constituents may require use of push-pull systems; use of cyanides require stringent precautions with the use of acids (possible HCN formation); general HVAC system can cause crossdrafts that will dilute air contaminants into the general plating room air.

Diffusion Furnaces. Local exhaust provided at "source end" jungle, and at "load end" of tube, using collector-type end caps to route tube exhaust products to vestibule exhaust opening; ensure VLF & HLF systems do not cause eddy currents and transient leaks/odors.

Chemical Storage Cabinets. Minimum of 50 CFM per standard sized cabinet; ensure compatibility of materials stored within cabinet; use rated flammable storage vessels if flammables are transferred from original container.

Equipment Cleaning Hoods. Used for cleaning equipment parts that are contaminated with arsenic, antimony or spinner/coater residues, and pumps, etc.; typically want a minimum face velocity of 125 lf/min at hood face; housekeeping in the hood is very important; also important to ensure the hoods are big enough to handle the largest parts to be cleaned.

Table 4.4. (Cont'd)

Pump & Equipment Exhaust Lines. Chemical compatibility with effluents, ability to handle pyrophoric materials, and potential for duct fires; blockage of ducting is possible from reaction products (such as CVD products from Si_3N_4); monitor exhaust duct/line pressure closely.

Portable Chemical Hoods. Use with non-flammable materials or as temporary exhausts for equipment or materials that are cleaned sporadically; they require preventative maintenance more than in-place chemical hoods and are more limited in the chemicals that can be effectively used in them.

Glove Boxes. Used for the containment of highly toxic source materials in vials or jars (antimony, arsenic) or flammable metals (phosphorus); need LEV to provide negative pressure at load door, and HEPA filtration on LEV duct to capture fugitive particulate.

Burn-In Testers. Temperature testing for IC's; need LEV for heat, and smoke detectors for IC's or boards that may start to decompose from dropping on heating elements or over-temperature shutoff malfunctioning on tester.

Drying Ovens including Blue M®-type Ovens. Used primarily to drive-off moisture after cleaning of equipment parts; other volatiles can come off IC package materials and cause odors if not purged properly prior to door opening; sometimes due to temperature control problems, local exhaust is provided via a exhaust duct damper opened immediately prior to door opening; sometimes exhausted for heat only.

Welding/Brazing/Cutting Hoods. LEV control velocities vary depending on the composition of the materials being worked on; LEV exhaust duct must be situated to avoid re-entrainment into general HVAC system.

Abrasive Blasting Hoods. Commonly used for cleaning contaminated parts from metal deposition areas (copper, platinum, gold, titanium/tungsten, aluminum, etc.); segregate "bead blasters" that are used for cleaning ion implanter parts (arsenic-contaminated or phosphorous-contaminated—which is a fire hazard); and other metallization equipment; ensure fine particle leakage is contained within the bead blaster by closing leakage points; establish controls/procedures for changing the contaminated bead blasting media (usually glass bead or silicon carbide).

Table 4.4. (Cont'd)

HEPA Portable and House Vacuum Systems. Portable HEPA vacuums are used extensively for cleaning the interiors of ion implanters or collector deposition in diffusion furnaces, cleaning-up residues in parts-cleaning hoods or around bead blasters; care must be taken in using the standard HEPA vacuum without impregnated activated charcoal (designed for vacuuming up mercury) as arsenic vapors or arsine gas may be generated during implanter and molecular beam epitaxy—the impregnated activated charcoal was very effective in stripping out contaminants from the vacuum exhaust stream;[28] house vacuum systems are very susceptible to contamination with hazardous materials in the system; they should not be used to clean surfaces contaminated with hazardous materials; however, because their use is difficult to control and they must be treated as being contaminated (i.e., full protective equipment and procedures).

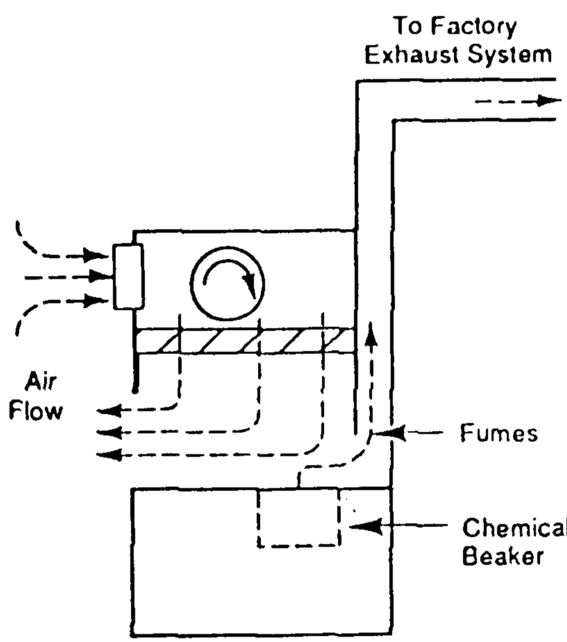

Figure 4.11. Cross section of a VLF fume exhaust hood illustrating the effect of the VLF on the hot etch pot LEV exhaust flow in a standard wet sink design.[27] *(Van Zant, P.,* Microchip Fabrication, *© 1990, McGraw-Hill, reprinted with permission.)*

Process Equipment. The majority of the process equipment used in semiconductor manufacturing requires the use of local exhaust ventilation for removing heat, particles, or chemical effluents. Some of the more critical process tools or processing steps that require the use of LEV systems are listed in Table 4.5.

Table 4.5. LEV Matrix for Semiconductor Fab Processes.[2]

PROCESS	EFFLUENT TYPE	SPECIAL NOTES
Photolithography		
• Cleaning • Photoresist Application • Drying/Baking • Mask Align/Exposure • Developing • PR stripping	• Acids • Solvents • Solvent/Heat • Heat • Solvents/Hydroxides • Solvents/Acids/Decomposition products	May require destruction of VOCs through incineration & metal or non-combustible ducting • Aromatic solvents: xylene • Acetates: n-Butyl Acetate • Glycol Ethers: EE, ME, EEA, MEA, PGMEA, ethyl lactate • Acids: Sulfuric primarily • Hydroxides: Sodium
Etching		
• Wet • Dry —Chemical —Physical —Reactive Ion	• Acids • Plasma etch gases & decomposition products	Requires use of wet scrubbers & chemically resistant ducting; residues in etch chamber. • Acids: hydrofluoric, sulfuric, phosphoric, nitric, hydrochloric, acetic and BOE (buffered oxide etch). • Gases: Freon®, oxygen, argon, and nitrogen. • Vacuum pump systems (oils, chlorinated/fluorinated residues) Note: Freon® gases are being phased out as *ozone depleting substances* (ODS), and wet scrubbers are ineffective in their removal.

(Continued)

Table 4.5. (Cont'd)

PROCESS	EFFLUENT TYPE	SPECIAL NOTES
Oxidation	• Water • Flammable gases	Flammable gas detection required for hydrogen gas.
Diffusion • Gas • Liquid • Solid	• Acid Gases • Corrosive Liquids • Toxic Metals	Requires use of exhaust gas conditioning systems; chemically and heat resistant ducting; continuous gas monitoring systems. Hydrogen chloride, phosphorus oxychloride, boron tribromide, antimony, boron, and phosphorus.
Ion Implantation	• Hydride Gases • Toxic Metals • Acid Gases • Hydrogen (as a carrier gas)	Requires use of chemically and heat resistant ducting; continuous gas detection systems. • Arsine, phosphine, diborane, boron trifluoride and arsenic pentafluoride. • vacuum pump systems (oils, decomposition products, cryo & turbomolecular regeneration products). Extensive use of lecture bottle quantities of pyrophoric/toxic/corrosive gas systems.

(Continued)

Table 4.5. (Cont'd)

PROCESS	EFFLUENT TYPE	SPECIAL NOTES
Epitaxial Deposition (including Molecular Beam - MBE)	• Acid Gases • Silicon Containing Gases • Hydride Gases • Hydrogen (as a carrier gas)	Requires use of exhaust gas conditioning systems; chemically and heat resistant ducting; continuous gas monitoring systems • HCl, dichlorosilane, trichlorosilane, silicon tetrachloride, arsine, phosphine and diborane. Extensive use of vacuum pump systems (oils, decomposition products). Extensive use of small cylinder quantities of pyrophoric/toxic/corrosive gas systems.
Chemical Vapor Deposition (CVD)	• Acid Gases • Silicon Containing Gases • Hydride gases • Pyrophorics • Hydrogen (as a carrier gas)	Requires the use of exhaust gas conditioning systems; chemically and heat resistant ducting; continuous gas monitoring systems. • HCl, phosphorus trichloride, silane, arsine, phosphine, diborane and dimethyl zinc. • extensive use of vacuum pump systems (oils, filters & traps, pyrophoric/toxic/corrosive gas hazards). • extensive use of high concentration pyrophoric/toxic/corrosive gas systems. Note: the use of dry pump systems is increasing; see Ch. 2, Sec. 4.1: Process: Vacuum.

(Continued)

Table 4.5. (Cont'd)

PROCESS	EFFLUENT TYPE	SPECIAL NOTES
Metallization • Thin Film E-Beam Sputtering	• Metals • Inert Gases	Use of vacuum pump systems. • Aluminum, copper, platinum, gold, silver and titanium/tungsten.
Metallization • Thick Film Plating	• Acids • Cyanides • Metals	Requires use of wet scrubber systems. • Sulfuric, phosphoric, nitric and acetic acids. • ferric or potassium/sodium cyanides. • Copper, gold, platinum and lead/tin.
Metallization • Alloying/Annealing	• Inert gases • Flammable Gases	May require use of flammable gas detection systems and metal exhaust ducting. • hydrogen, or hydrogen mixtures.

Design Criteria. The other area warranting mention in this overview of LEV systems is the element of design criteria for the system. Specifically, the materials of construction of the entire system (hood/sash, ducting, fan, exhaust conditioner, and roof level stacks) are extremely important. Many times, this selection for compatibility ends up being a trial and error process because the exact chemistry of the effluent of the process step or process tool is not precisely identified by the tool manufacturer, or exhaust conditioner manufacturer. It is fairly common (but becoming less of a problem because of tighter fire/building code compliance for compatibility of LEV system components with the effluent they are designed to exhaust) to find LEV exhaust ducts intermingled as they proceed from the process tool to the exhaust conditioning system. This co-mingling of process exhaust lines allows for reaction products and residues that are based on reaction dynamics of the process tools in use at the same time. This is not an area that has been studied extensively and evaluation usually comes from a specific problem or on-going exhaust/re-entrainment issue that has surfaced.

Materials of Construction. Based on the authors' personal experiences, any metal ducting or gas delivery tubing (including higher grades of stainless steel) that are exposed to even low levels of hydrogen chloride (HCl) gas and moisture will eventually become corroded enough to cause process contamination problems in the cleanroom, and require replacement. Recent information suggests that very high grades of stainless steel (SS 317L, Inconel® or Hastalloy®) may provide better protection against HCl corrosion (and allow for a single gas delivery line versus double contained lines under the Toxic Gas Ordinance). Another chemical that is notorious for attacking rubbers or elastomers in gaskets on equipment or ducting is n-methyl-2-pyrrolidone (M-Pyrol®). If M-Pyrol is to be used, gaskets will need to be made of Teflon® or Viton® elastomer.

The use of *aqua regia* (a mixture of red fuming or standard nitric acid and hydrochloric acid in a ratio of 1:3 that evolves pungent fumes/vapors of nitrogen oxides [NO_2 and N_2O_4], nitrosyl chloride, and possibly HCl) for the etching of layers of noble metals such as platinum and gold are infamous for corroding ducting/hoods/fans, and also creating emissions that are not easily scrubbed in wet scrubber systems. This difficulty in scrubbing aqua regia LEV exhaust effluents can lead to on-going re-entrainment issues into the general HVAC system. In some cases, a pretreatment neutralization system (that has pH sensing and interfaced hydroxide dispensing to adjust the air stream pH) is installed when highly corrosive aqua regia exhaust effluents are routed into a prescrubber. This adjusted air is then run through the normal wet scrubber exhaust conditioning system effectively scrubbing out the remaining aqua regia exhaust contaminants. Typically, the highest concentration of wet etch sink exhaust effluents occur when new solutions are poured in the etch baths, and during the draining of the etch basins by aspiration.

Fire Sprinklers. Another important design criteria is the installation of fire sprinklers in certain LEV system exhaust ducting.[29] Some fab exhaust ducts are exempt from the U.S. fire code requirements for in-duct sprinklers based on the materials of construction of the duct and interior diameter. U.S. fire codes [UFC Article 51 (c)(1)(D)(i)] require the use of either non-combustible ducting or in-duct fire sprinklers for ducts greater than 10 inches in diameter. Fire sprinklers are typically not used or located in the primary vertical runs in metal exhaust ducting or on systems running pyrophoric materials. This exemption of in-duct sprinklers is due to the potential for melting the fusible links in the sprinkler heads and flushing large quantities of fire sprinkler water on hot process tools.

Dampers. The use of dampers in the LEV system for balancing the individual hoods is another area of potential concern. The use of butterfly valves are notorious for being mismarked or installed incorrectly, and therefore misadjusted. Based on this experience, many companies have gone to the use of slider valves that physically indicate their relative position, and are easily installed with a set screw. This use of slider valves is recommended for all dampers. The primary problem with slider valves is that you must have physical clearance to allow the installation or removal of the slider. The compatibility of the damper with constituents in the exhaust stream is also an area that warrants evaluation.[30]

Exhaust Status Indicators. The complexity of the LEV systems used in a typical fab require an effective method of quickly determining whether the hood or exhaust duct is functioning properly. It is recommended that a prominently mounted aneroid-type gauge that directly identifies the static, velocity or total pressure with a Pitot tube be used.[30] For critical applications, an electronic aneroid gauge can be used with dry contacts that can drive a warning light or alarm, or notify a central control system. The most common examples of this type of gauges in use in the semiconductor industry are the Magnehelic® (aneroid-type) and Photohelic® (electronic aneroid-type) gauges. They should be calibrated initially, and on an annual basis. The gauge face should be marked for the normal pressure that is expected and lowest level that is safe. Note, any blockage between the static pressure tap and the hood/process tool would give a incorrect high reading with inadequate exhaust at the hood or tool. If a Pitot tube is to be mounted in the duct (for reading velocity pressure), ensure the environment does not contain corrosive gases or particles that will corrode or block the Pitot tube and give erroneous readings.

Gas Cabinets. Gas cabinet manufacturers use the Uniform Fire Code section (Article 51.107(c)) as the basic design criteria.[30] This section requires that gas cabinets be ventilated and that the average face velocity at the access port to the cabinet be not less than 200 FPM, with a minimum velocity at any point on the front plane of the access port of 150 FPM. For a standard configuration gas cabinet, you can use 150–200 CFM per bottle as the basic specification for the overall rough out of the required volumes for the various size cabinets. It is highly recommended that exhaust status indicators be specified for all gas cabinets. If highly toxic or pyrophoric compressed gas or liquid is to be used, a Photohelic-type gauge should be installed with a tie into warning alarms that output to the cylinder area. The output should also go to the site's central alarm control station. Fire code also requires that continuous gas monitoring detection be provided in the gas cabinet and the room housing the cabinet (unless open air).

2.3 Particulate Control Systems

The PCS is an integral part of semiconductor processing. A PCS acts as a subsystem for wafer isolation at the process tool, workstation, or zone of the fabrication area, or in the case of wafer pod carriers, as a mobile PCS for the wafers at critical processing steps. The basic idea of a PCS is to minimize the area that must be kept at these phenomenally low airborne particulate levels needed for class 10, 1, or 0.1 processing. The wafer is enclosed in an environment which is as small as possible, thereby isolating the wafer from contact with the fab room and human occupants. By controlling the number of particles in the air at locations where the loading and unloading of the wafers occurs, during the transport of wafers within the process tool or processing area, and at the process tool, this super-clean minienvironment can be established and maintained.

Standard Mechanical Interface/Wafer Isolation Technology. A standard has been developed that allows a standard mechanical interface (SMIF) on fab processing equipment and the PCS minienvironments. Newer terminology for PCS is known as *wafer isolation technology* (WIT). Some common definitions that are used in WIT were shown in Table 4.1. A basic *minienvironment system* includes three components: a tool enclosure, a cassette enclosure for transporting the wafers between tools, and an I/O port. Higher levels of particulate control require a third component or specialized method for loading the cassette enclosure into the tool. Conditioned air is typically supplied to the tool enclosure via self-contained filtration units or ceiling hung enclosures, through filtered ceiling ducts. Minienvironments, on the other hand, tend to isolate the process chemicals and potential emissions to the minienvironment. This hopefully isolates the fab worker from the exposure. The degree of isolation the minienvironment provides is a function of the air tightness of the enclosure, the supply/return/exhaust duct configurations, and the wafer input/output methodology. The concern from an exposure perspective comes during the maintenance activities on the minienvironment and interior equipment during process upsets that require direct access to the enclosure.[12][13]

This concept of minienvironments is very important from an industrial hygiene perspective, because the PCS can greatly affect the functioning of the LEV and general HVAC systems. Mention has already been made of the potential for eddy currents from the VLF's above a wet etch/clean system and at the load-end of a diffusion furnace. Some other examples include metal etchers when the bell jar is opened for wafer loading, and a HLF/VLF overpowers the LEV, and causes fugitive emissions from the residues on the interior surfaces of a metal etcher's bell jar. Another common occurrence is that of flow patterns established from a VLF above a processing station that utilizes hazardous materials. The return air loop below the equipment may capture fugitive emissions from the process and then recirculate them through the HEPA filtration system, which causes a significant dilution, but still perceptible odors and irritation to the operators of the process tool, adjacent tools, or processing bays. This is a very common transient odor situation, and can involve many hours of investigation to determine the exact location of the emissions, and recirculation path. Ultimately, the more isolated the PCS, the easier it is to determine the effect of the PCS on the LEV and general HVAC systems. The more intermingling of these system, the more difficult it is to pinpoint the source of odors or irritational smells in the fab. See Fig. 4.12 for a cross section of a VLF hood illustrating the recirculation flow on a standard workstation with a PCS above.[35]

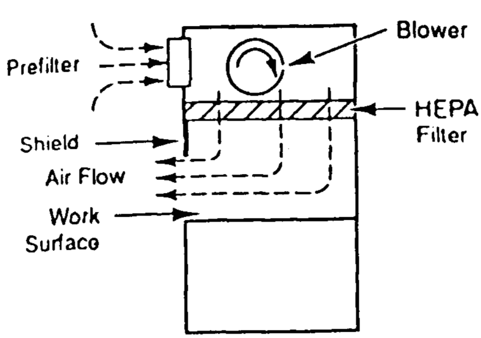

Figure 4.12. Cross section of a VLF hood illustrating the recirculation flow on a standard workstation with a PCS above.[35] *(Van Zant, P.,* Microchip Fabrication, © *1990, McGraw-Hill, reprinted with permission.)*

Body Suits/Rebreathers. Another PCS that is becoming in prominent use in newer Class 1 or 0.1 fabs is the use of total body suits for the fab area workers. In a sub-micron fab area, this PCS clothing involves the use of two separate particle isolating full body suits, gloves, foot coverings, and even supplied-air *rebreather systems* that supply HEPA filtered air through a portable belt-mounted pump system to the worker in the full body suits and helmets. As you can well imagine, these PCS suits can be confining, retain body heat if any strenuous activity is required and involve approximately 10 plus minutes of time to "suit up" prior to entering the fab area. This is an inconvenience for most fab operators, but a significant concern because of the time delay in ERT personnel exterior to the fab entering the fab area if the site requires ERT personnel to suit-up during a response. This effects the industrial hygienist in terms of emergency response planning, sampling methodology, and evaluation of exposure potential from emissions in the fab area. Typically, a separate set of precleaned industrial hygiene monitoring equipment is left in the sub-micron fab area for emergency or routine use. The prohibition against the use of paper and ink or pencil in sub-micron fab areas also makes note taking difficult. Tape recorders and/or portable computers have been used successfully for documenting events.

3.0 MONITORING

Ventilation systems are mechanical devices that require periodic maintenance and measurement of air volume, velocity, and contaminant loading. The measurement of the efficiency and effectiveness of LEV systems is a significant variable in the multiple layers of engineering controls that are utilized in the semiconductor hazard control strategy. Ventilation monitoring and correction of exhaust deficiencies identified is a critical activity that is many times overlooked until a serious problem develops. After a building or fab area has been built or modified, especially if the use of the building has changed or its HVAC system has been modified, HVAC testing, adjusting, and balancing (TAB) are extremely important aspects in the delineation and correction of ventilation-related air quality issues.[36]-[45]

3.1 Types

LEV Static Pressure Sensing. This type of gauge is standard equipment in most fab areas for exhausted process equipment. These devices measure the static pressure in exhaust ducts, which acts as an indicator of exhaust flow. The gauges should be prominently mounted on the exterior of the equipment enclosure or work bay. They should be marked to indicate both normal and minimum required static pressures on the exhausted system. This marking will allow equipment operators to quickly identify deficiencies in the ventilation at their specific process tool that may result in chemical exposures and/or process problems. Finally, the device should allow emergency response personnel to quickly verify the adequacy of the relevant local exhaust when responding to toxic gas monitor alarms or odor complaints. Sometimes the devices are used to trigger an alarm and automatic shutdown of the equipment when the static pressure drops below a predetermined set-point (electronic aneroid-type gauge, e.g., Photohelic®). Figure 4.13 illustrates the most popular form of an aneroid gauge for sensing static pressure—the Magnehelic® gauge. The principal advantages of this gauge are: easy to read, greater response than manometer types, very portable (small physical size and weight), absence of fluid which means less maintenance, and mounting/use in any position is possible without loss of accuracy. The principal disadvantages are: the gauge is subject to mechanical failure, requires periodic calibration checks (recommended annually), and occasional recalibration.[30] Also, unlike manometers which are primary standards, magnehelics are secondary standards.

LEV Actual Flow Sensing. Devices such as a Pitot tube with an inclined manometer or aneroid gauge, swinging or rotating vane anemometer, or "hot wire" electronic flow detectors are examples of actual flow sensing devices. Table 4.6 provides a summary of the characteristics of flow sensing instruments.[29] These devices are more sophisticated, and therefore are more expensive. They also require periodic maintenance and calibration to ensure accuracy. The advantage of this category of sensor is that it is actually detecting the velocity of air movement versus static pressure (or sucking potential). An example where this distinction is important is when a LEV duct may actually be clogged or blocked downstream of the static pressure sensor, and therefore be showing higher static pressure value on the manometer or Magnehelic®, but in fact be pulling little or no exhaust velocity through the duct.[30][38][41][42] Actual flow sensing devices come in direct contact with the exhaust stream, and are not recommended on exhaust systems handling corrosive materials.

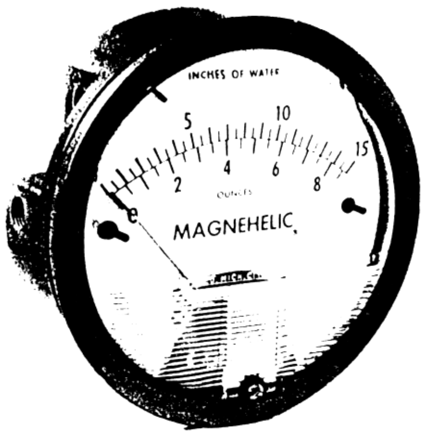

Figure 4.13. The most popular form of an aneroid gauge for sensing static pressure–the Magnehelic® gauge.[30] *(© 1992, ACGIH, reprinted with permission.)*

Because of the cost and maintenance associated with actual flow sensing devices, they are not routinely used in fab areas, with manometers and magnehelics being the sensors of choice. The QA/QC department typically has a calibrated velometer for determining the flow velocities on clean hoods. Most environmental health and safety groups also have both rotating vane and "hot wire" electronic air velocity instruments.

Practical Tips. Neither of the two categories of sensors already mentioned tell the semiconductor industrial hygienist if the exhaust system is effectively and efficiently capturing the air contaminants of concern. This requires the actual testing of the exhaust velocities at the point of capture at the LEV system, or at the "face" or opening of the hood, wet sink, or gas cabinet. This requires the services of a ventilation engineer, industrial hygienist, or technician under direction of an engineer or industrial hygienist. Ventilation monitoring is generally recommended to be performed on an annual basis, or more frequently if no exhaust indicator such as a magnehelic gauge (marked for appropriate ventilation) is provided. In either case, the monitoring data recorded should be put in a hardcopy form, and entered into a database or spreadsheet that allows the tracking of LEV system performance. Standardized monitoring protocols and record keeping systems are used commonly by the larger semiconductor manufacturers. Typically, face velocity readings are taken at the sash of lab hoods, entry point of the vestibules of diffusion furnaces, access ports on gas cabinets, glove

Table 4.6. Characteristics of Flow Sensing Instruments.[30] *(© 1993, ACGIH, reprinted with permission.)*

Instrument	Range (fpm)	Hole Size (for ducts)	Range, Temp.*	Dust, Fume Difficulty	Calibration Requirements	Rugged-ness	General Usefullness and Comments
Pitot Tubes with	600 - up	3/8"	Wide	Some	None	Good	
Inclined manometer							
Standard	600 - up	3/8"	Wide	Some	None	Good	Good except at low velocities
Small Size	600 - up	3/16"	Wide	Yes	Once	Good	Good except at low velocities
Double	500 - up	1/4"	Wide	Small	Once	Good	Special
Swinging Vane Anemometers							
	25-10,000	1/2-1"	Medium	Some	Frequent	Fair	Good
Rotating Vane Anemometers							
Mechanical	30-10,000	Not for duct use	Narrow	Yes	Frequent	Poor	Special; limited use
Electronic	25-200 25-500 25-2,000 25-5,000	Not for duct use	Narrow	Yes	Frequent	Poor	Special; can record; direct reading

*Temperature range: Narrow, 20-150 F; Medium, 20-300 F; Wide, 0-800 F.

boxes, and at the shroud of wet etch sinks. A "hot wire" anemometer or swinging vane velometer is normally used. Sashes on hoods are marked to indicate the sash position to achieve the minimum acceptable flow rate on the hood, and magnehelics or manometers marked to indicate acceptable static pressure on that exhaust system of a process tool.[36][38][43]

It is important to ensure that static pressure taps are installed between the hood or process tool and flow adjustor such as dampers.

Two other techniques of LEV, general HVAC and PCS ventilation monitoring that are qualitative versus quantitative are the use of dry ice, liquid nitrogen, or a water vapor wand for visual air flow pattern observation and tracer gas studies typically using sulfur hexafluoride (SF_6) for cross-contamination testing of the three ventilation systems.[32]-[34][46]

Note: because sulfur hexafluoride is an *ozone-depleting substance* (ODS), it is anticipated that production will be phased out in the near future.

3.2 Dry Ice, Liquid Nitrogen, and Water Vapor Wand

The standard IH technique for visually evaluating the effectiveness of a LEV system is through the use of particle generating *smoke tubes* that liberates titanium tetrachloride or other irritative smoke particulates that can be visually observed. The particle control requirements for fab areas obviously preclude the use of fine particle generating techniques in the fab. The use of carbon dioxide in the form of dry ice, liquid nitrogen from a dewar, or a water vapor generator with a wand work very effectively to allow the tester to see the actual flow patterns around the process tool or equipment. The competition for air is apparent as the different pressure gradients of the LEV, general HVAC, and PCS interact. This technique is always quite enlightening. It allows you to see the actual flow patterns at a piece of equipment or hood, and the mitigating effects on the LEV system. Dry ice, liquid nitrogen, or water vapor can be used for qualitatively identifying the location of continuous gas monitoring system sampling lines to ensure that the point is effective in picking up air contaminants from their generation point.[33][34]

3.3 Tracer Gas (SF_6)

The use of a tracer gas for evaluating the effectiveness of a LEV system, and the potential for the exhaust emissions being re-entrained back into the general HVAC system are well documented in the industrial hygiene literature.[47]-[49] This evaluation technique requires the use of an analytical detector that is sensitive to sulfur hexafluoride (SF_6) at the parts per billion level. Typically, a infrared detector or gas chromatograph with an *electron capture detector* (ECD) system is used. The standard scenario for the use of tracer gas is when the fab area has re-entrainment problems and on-going evacuations/concerns. The tracer gas is released through an LEV duct from the suspected piece of equipment, and the detector is set up on the general HVAC system that has been the source of the transient odors. The objective of the testing is to see if the exhausted air contaminant can be short-circuiting into the intake of the HVAC system (re-entrainment), and fed back into the fab area. Figure 4.14 provides a schematic representation of exhausted emissions being sampled for re-entrainment in the general HVAC system.[46] Tracer gas testing is a very powerful technique for use by the industrial hygienist to "prove" that re-entrainment is or can be occurring. [32]-[34][46] As was noted previously, sulfur hexafluoride is an ozone depleting substance, and it's production is in the process of being phased out.

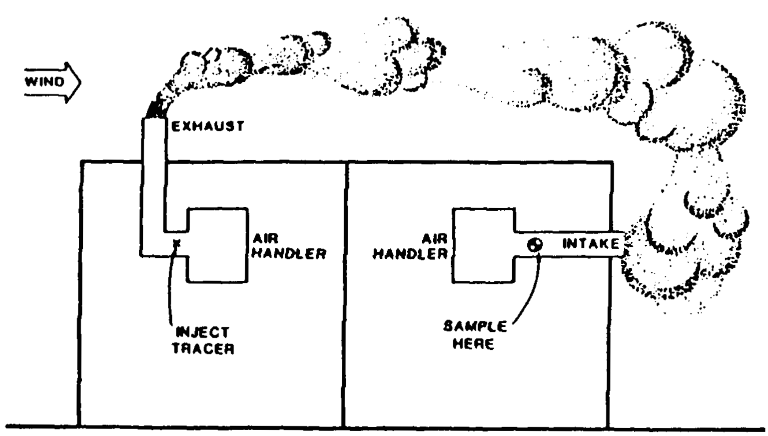

Figure 4.14. Schematic of exhausted emissions being sampled for re-entrainment in general HVAC system.[46] *(© 1989, ACGIH, reprinted with permission.)*

4.0 GUIDELINES/STANDARDS

There are numerous standard-setting groups and regulatory agencies that provide ventilation-related guidance. In the standards-setting groups, ASHRAE's Standard 62-1989: Ventilation for Acceptable Indoor Air Quality [14] and ASHRAE's Standard 55-1992: Thermal Environmental Conditions for Human Occupancy[39] are quite comprehensive. ASTM has Standard E741-83, Standard Test Method for Determining Air Leakage Rate by Tracer Dilution.[50] Another important standard-setting group is the American Conference of Governmental Industrial Hygienist (ACGIH), and their publication: *Industrial Ventilation, A Manual of Recommended Practice,*[16] is the "bible" for most practicing industrial hygienists. The National Institute of Occupational Safety and Health (NIOSH) also has published numerous documents on LEV systems and testing

Federal and State OSHAs (such as CAL/OSHA) have extensive standards for LEV systems on open surface tanks, laboratory-type hoods, and metals-working operations. Finally, the uniform fire, building, and mechanical codes have extensive standards on many ventilation-related items, including gas cabinets and other systems.

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